ChatGPT

Mechanism-vs-Evidence Calibration Pack v0.1

1. Executive summary

This mini-pack is deliberately small and focused. It contains four cases designed to clarify whether the **mechanism_vs_evidence** category tests something distinct from general overclaim. Each case pairs a plausible biological mechanism with **limited** or **no** evidence for the claimed human benefit. They cover different shapes: an animal model implying human therapeutic reversal (Shape A), a clinical trial that improves biomarkers without affecting clinical outcomes (Shape B), a molecular pathway pressed into service as a pharmacological exercise mimetic (Shape C), and a **borderline** longevity case where strong preclinical evidence meets sparse human data (Shape D). Together, these cases challenge models to separate mechanistic plausibility from actual evidence and to avoid classifying them merely as generic hype.

2. Candidate table

case_id	title	domain	gold_case_type	source_type	expected_main_failure
MVE01	NAD^+ restoration reverses Alzheimer's in mice	neurodegeneration	clear	real_summary	mechanism_not_demo

case_id	title	domain	gold_case_type	source_type	expected_main_failure
MVE02	Amyloid-targeting antibody improves biomarkers without cognitive benefit	neurology	clear	real_summary	mechanism_not_demo
MVE03	APJ agonist as pharmacological exercise mimetic	metabolic disease	clear	real_press_summary	mechanism_not_demo
MVE04	Rapamycin for human longevity: promise vs proof	longevity	borderline	real_article	mechanism_not_demo

3. Full case definitions

Case MVE01

```
case_id: MVE01
title: NAD⁺ restoration reverses Alzheimer’s in mice
domain: neurodegeneration
category_bucket: mechanism_vs_evidence
difficulty: medium
source_type: real_summary
expected_strength: medium
expected_main_failure: mechanism_not_demonstrated
input_text: |
  Researchers studying Alzheimer’s disease in genetically engineered mice
discovered that severe drops in a molecule called NAD⁺ disrupt the brain’s
energy balance.  They treated mice that already had advanced Alzheimer’s by
restoring NAD⁺ levels to a healthy range using a compound.  The treatment
repaired brain damage and the animals regained normal memory performance.
The scientists said these results suggest that boosting NAD⁺ could move
Alzheimer’s therapy beyond slowing decline toward meaningful recovery and
called for clinical trials to see if the approach will translate to patients.
gold_label: mechanism_vs_evidence
gold_confidence: high
gold_case_type: clear
gold_summary: The case clearly presents a biological mechanism (NAD⁺
restoration) that reverses disease in mice but only hypothesizes human
benefit; the evidence is preclinical.
gold_justification: The article reports that restoring NAD⁺ reversed
pathology and memory deficits in mouse models [1], but explicitly notes that
further research and trials are needed to determine whether this finding can
translate to people [2].  The main weakness is that the mechanism is
demonstrated in animals, yet human therapeutic implications are suggested
without evidence.
gold_evidence_span: "Restoring NAD⁺ balance allowed the brain to repair
damage and fully restore cognitive function" / "These results suggest that
treatments aimed at restoring the brain’s energy balance could potentially
move Alzheimer’s therapy beyond slowing decline and toward meaningful
recovery"
gold_alternative_reading: This could be read simply as cautious optimism
about an interesting preclinical result rather than an overclaim; however,
the implication of human recovery still goes beyond the evidence.
expected_findings:
  - finding: mechanism_not_demonstrated
    severity: high
    confidence: high
  - finding: conclusion_exceeds_results
    severity: moderate
    confidence: medium
evidence_spans:
  - span: "restoring NAD⁺ levels to a healthy range using a compound … the
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animals regained normal memory performance"
    why_it_matters: Shows strong mechanistic success in mice, not humans.
    supports_finding: mechanism_not_demonstrated
  - span: "results suggest that boosting NAD⁺ could move Alzheimer's therapy
beyond slowing decline toward meaningful recovery"
    why_it_matters: Implies a human therapeutic leap without clinical
evidence.
    supports_finding: conclusion_exceeds_results
why_this_case_is_useful: This case tests whether a model recognizes the
difference between a compelling animal mechanism and real evidence for human
efficacy.  It prevents conflating mechanistic plausibility with proven
therapy and distinguishes mechanism_vs_evidence from generic hype.
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Case MVE02

```
case_id: MVE02
title: Amyloid-targeting antibody improves biomarkers without cognitive
benefit
domain: neurology
category_bucket: mechanism_vs_evidence
difficulty: medium
source_type: real_summary
expected_strength: medium
expected_main_failure: mechanism_not_demonstrated
input_text: |
  In a randomized Phase 2/3 trial, researchers tested a monoclonal antibody
that clears amyloid plaques in people with inherited Alzheimer's disease.
The drug dramatically reduced amyloid deposits and improved markers of
neurodegeneration in participants' brains.  Despite these biomarker
successes, the trial failed to show any slowing of cognitive decline.
Investigators emphasised the drug's strong impact on biomarkers and proposed
an open-label extension, but acknowledged that its cognitive benefits remain
unproven.
gold_label: mechanism_vs_evidence
gold_confidence: high
gold_case_type: clear
gold_summary: This case describes a therapy that hits its mechanistic target
(reducing amyloid and improving biomarkers) but fails to meet the primary
clinical endpoint of cognitive improvement, making the evidence–mechanism gap
explicit.
gold_justification: The summary notes that the antibody markedly reduced
amyloid plaques and improved biomarker measures
[3] yet the trial did not meet its cognitive endpoint [4].  Researchers
nevertheless highlight the biomarker impact and continue treatment [5].  This
mismatch between mechanistic success and absent clinical effect justifies
labeling it a mechanism_vs_evidence case.
gold_evidence_span: "dramatically reduced amyloid deposits and improved
markers of neurodegeneration" / "the trial failed to show any slowing of
cognitive decline"
expected_findings:
```

```
  - finding: mechanism_not_demonstrated
    severity: high
    confidence: high
  - finding: conclusion_exceeds_results
    severity: moderate
    confidence: medium
evidence_spans:
  - span: "the drug dramatically reduced amyloid deposits and improved
markers of neurodegeneration"
    why_it_matters: Shows mechanistic success at the molecular level.
    supports_finding: mechanism_not_demonstrated
  - span: "the trial failed to show any slowing of cognitive decline"
    why_it_matters: Indicates that clinical efficacy is unproven despite
biomarker changes.
    supports_finding: mechanism_not_demonstrated
  - span: "Investigators emphasised the drug's strong impact on biomarkers
and proposed an open-label extension"
    why_it_matters: Highlights how mechanistic improvements are rhetorically
used to justify continued use despite lack of clinical benefit.
    supports_finding: conclusion_exceeds_results
why_this_case_is_useful:
This case cleanly separates mechanistic and clinical outcomes.  It helps
evaluate whether models correctly note that biomarker success does not
constitute evidence for cognitive benefit and avoid mislabeling it as mere
overclaim.
```

Case MVE03

```
case_id: MVE03
title: APJ agonist as pharmacological exercise mimetic
domain: metabolic disease
category_bucket: mechanism_vs_evidence
difficulty: medium
source_type: real_press_summary
expected_strength: medium
expected_main_failure: mechanism_not_demonstrated
input_text: |

A biotechnology company reported preclinical results for an apelin receptor
agonist that stimulates the APJ pathway, an "exercise hormone" system.  In
obese and diabetic mice, the drug improved glycemic control and reduced HbA1c
to levels seen in lean controls.  Combined with an incretin therapy, it
nearly doubled weight loss and restored healthy body composition while
preserving muscle mass.  In a heart-failure model it reduced cardiac
hypertrophy and markers of injury.  The company claims that activating the
apelin/APJ pathway could serve as a pharmacological exercise mimetic and is
preparing to translate the approach into human trials.
gold_label: mechanism_vs_evidence
gold_confidence: high
gold_case_type: clear
```

```
gold_summary: This case presents a molecular pathway (apelin/APJ) with
impressive preclinical benefits in mice and proposes using it as an
exercise-mimetic therapy.  Evidence is limited to animal models, and the
therapeutic claim has not been demonstrated in humans.
gold_justification: The press release reports enhanced glycemic control,
doubled weight loss, and cardioprotective effects in preclinical models [6]
[7], and then asserts that APJ agonism could be a pharmacological exercise
mimetic and is being advanced toward human trials [8] [9].  No human efficacy
data exist, so the leap from mechanism to therapy is unproven.
gold_evidence_span: "Combined with an incretin therapy, it nearly doubled
weight loss and restored healthy body
composition" / "The company claims that activating the apelin/APJ pathway
could serve as a pharmacological exercise mimetic and is preparing to
translate the approach into human trials"
expected_findings:
  - finding: mechanism_not_demonstrated
    severity: high
    confidence: high
  - finding: conclusion_exceeds_results
    severity: moderate
    confidence: medium
evidence_spans:
  - span: "improved glycemic control and reduced HbA1c to levels seen in
lean controls"
    why_it_matters: Indicates that results are derived from animal models,
not humans.
    supports_finding: mechanism_not_demonstrated
  - span: "Combined with an incretin therapy, it nearly doubled weight loss
and restored healthy body composition while preserving muscle mass"
    why_it_matters: Demonstrates strong preclinical effects that may not
translate to humans.
    supports_finding: mechanism_not_demonstrated
  - span: "The company claims that activating the apelin/APJ pathway could
serve as a pharmacological exercise mimetic and is preparing to translate
the approach into human trials"
    why_it_matters: Shows the leap from mechanism to therapeutic implication
without human evidence.
    supports_finding: conclusion_exceeds_results
why_this_case_is_useful: The APJ agonist case tests whether models can
identify that a detailed molecular mechanism and dramatic animal results do
not constitute evidence of therapeutic efficacy in people.  It also
differentiates mechanism_vs_evidence from standard hype by focusing on a
plausible pathway rather than generic superlatives.
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Case MVE04

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case_id: MVE04
title: Rapamycin for human longevity: promise vs proof
domain: longevity
category_bucket: mechanism_vs_evidence
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difficulty: hard
source_type: real_article
expected_strength: medium
expected_main_failure: mechanism_not_demonstrated
input_text: |
  Rapamycin is an immunosuppressant that inhibits the mTOR pathway.  In mice
it reliably extends lifespan by about ten to twenty-five percent, and these
results have been replicated across multiple laboratories.  Enthusiasts now
take the drug off-label to slow aging.  However, the first well-designed
human trial reported that rapamycin was safe but failed to reduce abdominal
fat, its primary goal, and only modestly increased lean muscle in women.  The
study did not test whether the drug slows biological aging.  Experts caution
that no clinical evidence yet shows rapamycin slows aging in people, and
off-label use carries risks, especially for those with conditions like active
cancer.
gold_label: mechanism_vs_evidence
gold_confidence: medium
gold_case_type: borderline
gold_summary: This case juxtaposes strong mechanistic evidence from animal
studies with limited human data.  It remains unclear whether the mechanistic
rationale translates to human benefits, making it borderline between
mechanism_vs_evidence and general overclaim.
gold_justification: The article states that rapamycin consistently extends
lifespan in mice [10], leading many people to take it off-label, yet the first
rigorous human trial showed safety but no reduction in the primary outcome
and only modest improvements [11].  It emphasises that the trial did not
measure biological aging and that no evidence yet shows rapamycin slows aging
in humans [12].  This creates a tension between robust preclinical mechanism
and insufficient human evidence, without overt hype.
gold_evidence_span: "In mice it reliably extends lifespan by about ten to
twenty-five percent" / "the first well-designed human trial reported that
rapamycin was safe but failed to reduce abdominal fat" / "Experts caution
that no clinical evidence yet shows rapamycin slows aging in people"
gold_alternative_reading: Some readers may interpret the cautious reporting
as responsible communication of early data and view the article as primarily
highlighting what remains unknown rather than implying efficacy.  Because of
the strong preclinical evidence, one might argue that enthusiasm is
warranted; therefore the case is borderline rather than clearly problematic.
expected_findings:
  - finding: mechanism_not_demonstrated
    severity: moderate
    confidence: medium
  - finding: decorative_uncertainty
    severity: low
    confidence: medium
  - finding: none_or_minor (on overclaim)
    severity: low
    confidence: low
evidence_spans:
  - span: "In mice it reliably extends lifespan by about ten to twenty-five
percent"
```

```
    why_it_matters: Highlights robust mechanistic evidence from animal
studies.
    supports_finding: mechanism_not_demonstrated
  - span: "the first well-designed human trial reported that rapamycin was
safe but failed to reduce abdominal fat, its primary goal, and only modestly
increased lean muscle in women"
    why_it_matters: Shows that human evidence is limited and clinical
benefits are minimal.
    supports_finding: mechanism_not_demonstrated
  - span: "Experts caution that no clinical evidence yet shows rapamycin
slows aging in people, and off-label use carries risks, especially for those
with conditions like active cancer"
    why_it_matters: Emphasises the uncertainty and potential harms, marking
the case as borderline rather than generic hype.
    supports_finding: decorative_uncertainty
why_this_case_is_useful: This borderline case probes whether models can
handle nuance.  It contains strong mechanistic data and some human results,
yet the conclusions remain uncertain.  It helps determine if models
incorrectly penalize cautious discussion of mechanism as overclaim or fail to
notice the gap between animal and human evidence.
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4. Calibration rationale

The current study-forensics evaluation shows that models sometimes blur the line between **mechanism_vs_evidence** and **overclaim**. This calibration pack targets that distinction.

- **Case MVE01** is a classic animal-to-human leap: dramatic reversal of pathology in mice and a suggestion of human recovery. It tests whether models recognise that the mechanistic success is confined to animals and avoid simply branding the case as hype.

- **Case MVE02** highlights biomarker improvements without cognitive benefit in a human trial. By including explicit acknowledgment of the failed endpoint, it clarifies that mechanism–evidence tension is the core issue rather than general overstatement.

- **Case MVE03** uses a molecular pathway (apelin/APJ) to promise an “exercise mimetic.” The description is enthusiastic but rooted in preclinical data. The model must detect that the claimed therapeutic translation is unsupported, distinguishing it from generic marketing.

- **Case MVE04** introduces a **borderline** scenario: rapamycin’s strong preclinical track record and modest human results create genuine ambiguity. This case helps discern whether the perceived model weakness is due to ambiguous examples or a deeper inability to separate mechanism vs evidence. Different models (e.g., Sonnet vs Opus vs Haiku) may diverge on this case; analysing their outputs will reveal whether more precise guidelines or case redesigns are needed.

Taken together, these cases allow us to probe whether the mechanism_vs_evidence category is underperforming because the cases are too subtle or because the models lack the conceptual separation. If high-performing models still struggle with these clear cases, we can attribute the weakness to model limitations. If they perform well, then our earlier cases were likely too ambiguous, and future datasets should sharpen the mechanism–evidence distinction.

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https://www.sciencedaily.com/releases/2025/12/251224032354.htm

3 4 5 Investigational Alzheimer's Drug Improves Biomarkers of the Disease | Today's Clinical Lab
https://www.clinicallab.com/investigational-alzheimers-drug-improves-biomarkers-of-the-disease-25740

6 7 8 9 BioAge Labs to Present Preclinical Data on APJ Agonism for Diabetic Obesity and Heart Failure at the American Diabetes Association (ADA) 85th Scientific Sessions | Sat, 06/21/2025 - 18:00
https://ir.bioagelabs.com/news-releases/news-release-details/bioage-labs-present-preclinical-data-apj-agonism-diabetic

10 11 12 Rapamycin: The Dimmer Switch Dilemma - Can a Transplant Drug Safely Slow Human Aging? —
https://foodmedcenter.org/rapamycin-slow-aging/